

Initial Test of MPU6050 Sensors

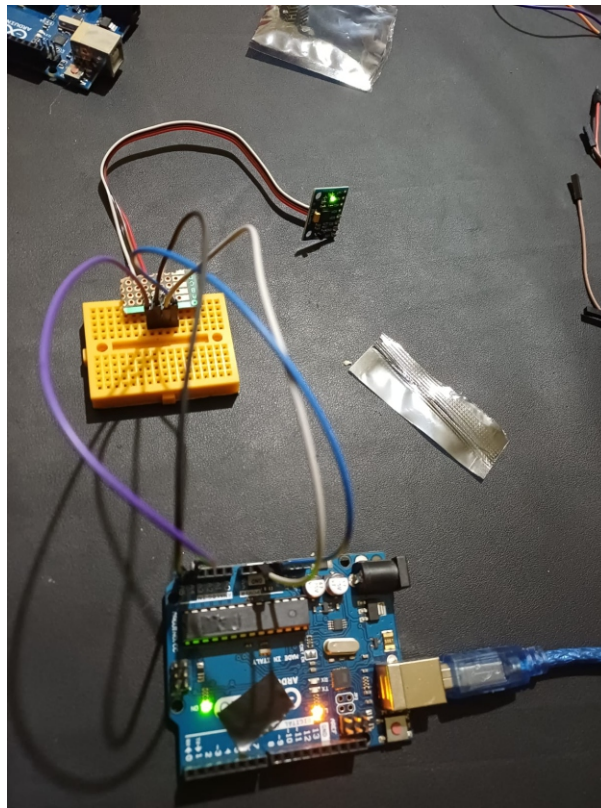


Figure 01: Testing of the MPU6050 sensors

Initial Test Code MPU6050 Sensors:

```
#INCLUDE <WIRE.H>
#include <MPU6050_LIGHT.H>

MPU6050 MPU(WIRE);

VOID SETUP() {
  SERIAL.BEGIN(115200);
  WIRE.BEGIN();

  BYTE STATUS = MPU.BEGIN();
  SERIAL.PRINT("MPU6050 STATUS: ");
  SERIAL.PRINTLN(STATUS);

  IF (STATUS != 0) {
    SERIAL.PRINTLN("COULD NOT CONNECT TO MPU6050! CHECK WIRING.");
    WHILE (TRUE); // HALT
  }

  SERIAL.PRINTLN("CALIBRATING... KEEP SENSOR STILL.");
  DELAY(1000);
  MPU.CALCOFFSETS(TRUE, TRUE);
  SERIAL.PRINTLN("DONE!");
}
```

```
VOID LOOP() {  
    MPU.UPDATE();  
  
    SERIAL.PRINT("ACCEL X: "); SERIAL.PRINT(MPU.GETACCX());  
    SERIAL.PRINT(" Y: ");      SERIAL.PRINT(MPU.GETACCY());  
    SERIAL.PRINT(" Z: ");      SERIAL.PRINTLN(MPU.GETACCZ());  
  
    SERIAL.PRINT("GYRO X: "); SERIAL.PRINT(MPU.GETGYROX());  
    SERIAL.PRINT(" Y: ");      SERIAL.PRINT(MPU.GETGYROY());  
    SERIAL.PRINT(" Z: ");      SERIAL.PRINTLN(MPU.GETGYROZ());  
  
    SERIAL.PRINT("ANGLE X: "); SERIAL.PRINT(MPU.GETANGLEX());  
    SERIAL.PRINT(" Y: ");      SERIAL.PRINT(MPU.GETANGLEY());  
    SERIAL.PRINT(" Z: ");      SERIAL.PRINTLN(MPU.GETANGLEZ());  
  
    SERIAL.PRINTLN("---");  
    DELAY(500);  
}
```